JOHANSSON SHIPYARD

Minato Maru - Engineering correction and independent review

JS-127-ER-900 | Revision B | 13 October 2026

Issue: FOR DESIGN REVIEW. Construction release withheld.

This report records implemented corrections to the Johansson World ferry and the accompanying dimensioned CAD reference. It separates verified numerical and game-geometry checks from unverified supplier interfaces and real-vessel construction requirements. The review is an AI-assisted computational review; it is not a survey, statutory certificate or signed professional construction approval.

Correction scope

- The hull loft at 1.900 m immersion displaces 375.41053 t in seawater, replacing the inconsistent 283 t claim. The retained 210 t lightship, 283 t operating target and 1.6 m motion GM remain explicit assumptions, not approved loading particulars.

- The former aft tank outside the hull is replaced by disjoint conformal double-bottom reservations under the auxiliary space. Tank envelopes, competing liquid inventories and a stated provisional structural deduction are coordinated numerically.

- Both generators move forward into the auxiliary room, clearing the main-engine fore-end service band. Labels, system terminals, local control points and cable route estimates follow the relocated equipment; walking checks use the actual clear aisles.

- Transformer and shore supply records distinguish primary and secondary voltage sides. Cable calculations include hot copper, DC return paths, grouped loads, protective conductors and star-delta tails. Emergency-generator motor-start capability remains a hold.

- Engineering route acceptance checks finite pipe radius, equipment, bounds and independent circuit interference. Decorative preview routes remain visibly separate from the engineering acceptance registry. Unresolved routes are scheduled individually.

- CAD model space uses millimetres and a controlled longitudinal/port/baseline datum. The PDF supplies physical A2/A3 sheets with measured scales; schematics and legacy SVG previews are marked NTS. Creator approval stamps are removed.

Source control

Audited repository snapshot: nj22az/nj22az.github.io, commit e87c06b16734d882626be7fe6cdd2ab153d649c4. Subsequent main commit 0aa7b46ac00cbf4aba6ba5d8bc049a820d5a154a retained identical hashes for the modified ferry source files. Corrections are identified separately in the source-change record. CAD generators reproduce the drawing issue from the saved JSON snapshot; drawing scales apply at 100% print only.

Review disposition

The package is a reviewed correction and design-reference issue. It does not establish that a real ship can be built from it. The independent review and outstanding-work register below govern the limits of acceptance; no programme of automatic checks replaces verification of hull structure, weights, stability, machinery supplier data, piping fabrication or classification requirements.

Independent completion review

Minato Maru - independent chief review IR1

Reviewed issue: CAD Revision B and the corrected local source based on commit e87c06b16734d882626be7fe6cdd2ab153d649c4, 13 October 2026.

Disposition: accepted as a checked correction and preliminary engineering-reference package for design review. **Construction release remains withheld.** This independent AI review is not a registered engineer's signature, survey, flag approval or classification approval. It does not describe a complete real-vessel building design.

Independent evidence

The reviewer reran every available tests/ferry-*.test.mjs test against the final local source: **80 passed, zero failed**, in 5.02 seconds. This includes generator relocation and centre/outer-aisle walking, longitudinal pipe bounds, engineering-route rejection, hydrostatic convergence/independent polygon integration, electrical side records and source-generated drawing consistency. Tests establish the model's behaviour and arithmetic; they do not prove real escape adequacy or physical-system performance.

The final drawing book contains **55 pages: nine A2 sheets and 46 A3 sheets**. All 55 final pages were visually inspected through seven montages; dense schedules and source ratings received closer inspection. Independent PDF inspection found correct paper sizes, Revision B, APPROVAL NONE and construction-withheld status on every page, with no off-page text words. The earlier table-note overlaps are closed. Measured PDF vector lengths confirm a 5 m scale bar is 50.000006 mm on the 1:100 context and 100.000006 mm on the 1:50 arrangement. The plan transform also gives 110 mm for the 5,500 mm engine-room length and 159.2 mm for 7,960 mm breadth.

Every one of the **55 sheet DXFs and the separate spatial DXF** was reopened independently with ezdxf and audited. All use millimetres; no errors or automatic fixes were reported. The section-sheet LINE entities now all lie in the XY drawing plane, closing the original mixed 2D/3D projection defect. The separate spatial model remains explicitly an equipment-envelope and pipe-centreline reference.

Live corrected ship particulars, GA, machinery, electrical particulars, loads, sources and feeders exactly match the final CAD source snapshot. The accepted engineering-pipe collection and unresolved registry also match. Final physical CAD uses **55 conceptually accepted paths**; **151 positive-DN connections remain unresolved**. Four additional topology links are signals. The **172 game previews remain unverified visual representations**, retained only as such. Accepted paths do not prove actual nozzle ownership, pipe classes, penetration integrity or a full-hull clash check.

Independent Python recalculation reproduced the machinery record without discrepancy: 24.12743158 L swept volume, 16.947054 bar BMEP, 9.00 m/s mean piston speed, 3,253.834 Nm crankshaft torque, 73.1031 L/crossing including assumed generator duty, and 9.2335 days on the crossing-only usable-fuel basis. Alongside operation, reserve policy and adverse conditions remain excluded.

The corrected one-line was checked against live records. Shore primary is 200 V, 144.3 A, 160 A frame/145 A provisional setting; ship-side secondary is 440 V, 65.6 A, 80 A frame/66 A provisional setting. ETR is 5 kVA at 440/105 V, with 6.6/27.5 A and 10/32 A frames, 7/28 A settings. The emergency set is correctly distinguished as 445 V nameplate and 440 V nominal distribution. Manufacturer selection, transient performance and protection coordination remain open.

Review findings closed before issue

Finding	Verified repair
Section DXF boundaries and equipment occupied different planes	Both use consistent 2D view projections; spatial model is separate
One-line retained old 75 A shore and 10 kVA ETR literals	Final page 50 is derived from side-specific corrected source records
Dense schedule rows overlapped verification headings	Line and feeder page chunks reduced; final pages inspected with clear note separation
Game previews could enter engineering CAD without acceptance	Physical export uses engineeringPipes; unresolved connections receive a separate register
Routing envelope lacked longitudinal bounds	Radius-aware longitudinal checks and regression pass; nozzle ownership limitation is explicit

Finding	Verified repair
Revision, physical scale and creator-approval metadata were inconsistent	Final B issue, actual mm scales, distinct manufacturer references and APPROVAL NONE verified

No remaining defect blocks issuing this bounded reference package. The following items still block real ship construction and physical acceptance.

Construction and physical-design holds

1. **Service and approval basis:** agree actual flag/class, domestic passenger/vehicle service, tonnage and applicable rules. The 1994 setting and nominal 199 GT are design narrative, not evidence of registration or certification.

2. **Weight and stability:** retained 1.900 m immersion gives 375.41053 t geometric seawater displacement. The conditional nominal 289.010 t schedule leaves 86.40053 t unexplained. The 210 t lightship is unverified; GM 1.6 m is a game-motion parameter. Resolve verified weights/CoG, trimmed hydrostatics, loading cases, GZ, downflooding, intact and damage stability. Arithmetic residuals are not payload approval.

3. **Structure and tanks:** the relocated paired trim tanks and extended main ballast allocation fit concept reserves: respectively 4.8524 m3 versus 4 m3 nominal and 16.4238 m3 versus 16 m3 nominal. ER fluid reservations of 16.9 m3 fit within 18.5526 m3 planning reserve. These values include stated geometric/structure allowances and are not fabricated net capacities. Plate/frame/foundation calculations, welds, partition/access/vent/suction arrangements and calibrated tank volumes remain missing.

4. **Actual machinery and shaft line:** the Johansson engine remains fictional; later OEM outlines and ratings are separate comparisons. Obtain selected engine/gear duty, certified mass/CoG, bolts/mounts/nozzles, foundations and thrust reactions, shaft alignment/bearings, propeller matching, fatigue and torsional vibration.

5. **Access and overhaul:** moving the generators into the auxiliary room corrects the demonstrated blocked service band and model-walk routes. Nominal room headroom remains 1.90 m versus a 2.00 m yard route target; the proposed engine envelope leaves 0.25 m overhead. Protected escape continuity, piston/liner/head removal, deck covers, lifting points, rigging, crane capacity/reach and lift stability remain unresolved.

6. **Piping and service duties:** resolve the 151 unsupported, missing or obstructed positive-DN connections. Conceptually accepted routes use planning radii and recorded game terminals. Actual pipe outside diameters/walls/insulation, fittings, supports, ownership, hull/fire/watertight penetrations, hydraulic losses and pump curves remain open. Ventilation, cooling, exhaust backpressure, fuel/air starting and fire-suppression protected volumes require selected-equipment calculations.

7. **Electrical installation:** complete actual generator/transformer/motor/cable selection, fault duties and discrimination, grouping/temperature installation checks, upstream-to-terminal drops, DC inventory and battery/endurance arrangements. The 52.5 kVA emergency fire-pump starting scenario exceeds the 40 kVA continuous rating; transient capacity is unknown. Current-only arithmetic cannot prove acceleration or voltage recovery.

8. **Final approval and acceptance:** qualified human review, selected suppliers, accepted class/flag drawings and applicable fabrication, shop, onboard and sea-trial testing are required. The package contains no such certification.

Scope and traceability

This review covers the supplied partial source snapshot, the generated CAD package and the listed engineering records. It does not establish a full town runtime build, browser performance, published-site deployment, as-built ship or real machinery trial. The 92,316-triangle/39-mesh regression is a software budget result, not a measured frame-rate guarantee.

The numerical/reference checks were informed by the independently opened primary ClassNK Part H June 2026 reference and Yanmar 2020 brochure. Applicability remains to be agreed. The legacy H physical reference and later C operating reference remain explicitly separate; outline literature does not supply a vessel-specific installation drawing.

Machine-readable evidence, reviewed artifact hashes, check counts, closed findings and hold IDs are in independent-review.json. The drawing-book SHA-256 is 58cdb31dee474ef41c7361196294cb159ed47df62fd31467c200068db2071d0d.

Hull, tank packing and conditional stability

Minato Maru - hull and stability engineering review

Revision basis: source snapshot e87c06b16734d882626be7fe6cdd2ab153d649c4, reviewed 13 October 2026. Calculation source: src/world/ferry-engineering-hydrostatics.js. Detailed inputs and results: hull-results.json.

Disposition: game hull inconsistencies corrected in the proposed particulars and tank allocation; real ship construction and stability release withheld. Numerical consistency of a game envelope is distinct from an approved vessel design. This review is a second calculation discipline within the project, not a signed naval architect or classification-society approval.

1. Exact hull, dimensions and the displacement defect

The calculation reproduces the actual parametric hull in ferry-model.js, rather than assuming that a block coefficient describes it. The source functions are sheer, deckEdge, halfBreadth, keelY, fullness, narrowing and sectionPoint. The original shipyard frame is x positive to port, y above baseline, z forward from the transom. The game origin is amidships on the design waterline. All calculations use metres, cubic metres and tonnes.

Quantity	Previous assertion	Calculated or controlled correction
Length overall	34.500 m	34.500 m retained
LBP reference	31.000 m	31.000 m retained as stated reference, not surveyed
Moulded breadth	8.400 m	8.400 m retained
Moulded depth / car deck	3.000 m	3.000 m retained
Design immersion	1.900 m	1.900 m retained to preserve physical propulsion arrangement
Immersed volume	Not calculated from hull	366.254176 m³
Displacement in seawater, density 1.025 t/m³	283.000 t	375.410531 t
Block coefficient on stated LBP	0.5600	0.7402663
Block coefficient on LOA	Not stated	0.6651669
Waterplane area	Not stated	259.2792 m²
LCB from transom	Not stated	19.25083 m
KB	Not stated	1.05678 m
Transverse BM	Not stated	3.89221 m
Transverse KM	Not stated	4.94898 m
TPC	Not stated	2.65761 t/cm
Lightship	210 t	Legacy assumed mass retained separately; not verified
Deadweight	73 t	Arithmetic difference 165.41053 t only; not certified capacity
Loaded GM	1.600 m	Animation assumption only; cannot be a verified stability particular
Gross tonnage	199 GT	Fictional service descriptor; tonnage is not displacement or a verified measurement

The old displacement differs from the current hull buoyancy by **92.41053 t, or 32.65%**. The corrected displacement is a calculation of the existing model at its chosen waterline. It is not proof that the real ship would weigh this amount. Neither lightship nor payload is increased to hide the difference.

2. Conditional immersion and propulsion consequences

Supplying the assumed 283 t weight gives an even-keel immersion of 1.521679992 m, and 210 t gives 1.198202014 m. These are conditional buoyancy solutions, not approved operating drafts. The 283 t case has KMt 5.46305 m; the 210 t case has KMt 6.42981 m.

The original propeller axis is physically 0.900 m above baseline, with 1.450 m nominal diameter. Its crown is 1.625 m above baseline and its bottom is 0.175 m above baseline. At the retained 1.900 m immersion the crown is 0.275 m underwater. At the conditional 283 t even-keel immersion the crown would be 0.10332 m above water, and at the 210 t even-keel immersion 0.42680 m above water. Stern-down trim can change these figures materially and must be solved before judging any actual loaded condition.

The model's old SY=-1.0 was a waterline-relative number. Changing the draft while keeping it would silently move the physical shaft relative to the keel. The parent design therefore retains the 1.900 m waterline and corrects the particulars, while a weight and trim reconciliation remains open. A propeller-immersion and shaft-line check is required for every eventual loading condition.

3. Invalid aft peak and accepted tank relocation

The original aft-peak GA box occupied z=0-2.6 m, x=+/-3.5 m and y=0-1.0 m. Its nominal box volume is 18.2 m³, but its intersection with the actual hull is **0.000 m³**. The counter's bottom is 1.95 m above baseline at the transom. The claimed 4 m³ ballast cannot be installed in that position.

An initial alternative under the engine room fitted as an isolated geometric tank. The integrated whole-room tank budget then rejected it: the ER double bottom provides only 18.55260 m³ after explicit reserves, whereas existing fluids plus another 4 m³ would require 20.9 m³. The correct action is to move the aft trim tanks into the auxiliary double bottom and extend the principal ballast allocation forward under the workshop/store footprint.

Tank zone	z range, m from transom	y range above baseline, m	Gross conformal envelope	After 10% planning deduction	Nominal target
ER storage fluids	8.5-14.0	0.10-0.68	20.61399 m³	18.55260 m³	16.9 m³
Paired aft trim ballast	14.1-15.4	0.10-0.68	5.39157 m³	4.85241 m³	2 x 2.0 m³
Paired main ballast	15.6-20.0	0.10-0.68	18.24865 m³	16.42379 m³	2 x 8.0 m³

The paired tanks start at x=+/-0.20 m. Their outboard sides follow the actual half-breadth minus 0.10 m. Their lower surface is max(0.10 m, hull bottom plus 0.10 m). The 0.10 m side/bottom reservation is a conceptual geometric clearance, **not a selected plating thickness**. The 10% deduction is a stated allowance for planning, **not calculated structure or a calibrated liquid capacity**. Whole forward DB envelope from z14-22.5 m is 35.24788 m³, with 31.72310 m³ after this allowance.

The original aft-peak source identifier may remain for compatibility, but its description and position must become the relocated aft trim DB tank, within fwd-double-bottom. Aft trim LCG is 14.75 m; main ballast LCG is 17.80 m. Tank partition, frame, manhole, sounding, vent, suction, drainage and neighbouring-service drawings remain required.

4. Explicit loading sensitivity

These scenarios deliberately disclose all masses and centres of gravity. Lightship is assumed at 210 t, KG=4.4 m, LCG=19.23 m. Passengers and crew use 90 kg/person including a baggage allowance; cars use 1.6 t each. The exact schedules are included in the JSON. The higher-KG case changes only assumed lightship KG to 5.4 m. Half-liquid cases retain full-state liquid CG heights as a conservative sensitivity assumption. Rectangular slack-tank dimensions are unverified bounding assumptions and do not define the actual tank construction.

Conditional case	Weight, t	Even-keel draft, m	KG, m	FSC, m	Corrected initial GM, m	Indicative bow-down trim, m	Initial heel, deg
legacy-lightship-only-assumed-KG4.4	210.000	1.1982	4.4000	0.0000	2.0298	0.061	0.00
full-nominal-stores-and-all-ballast	289.010	1.5474	3.8298	0.0000	1.5771	-0.799	0.00

Conditional case	Weight, t	Even-keel draft, m	KG, m	FSC, m	Corrected initial GM, m	Indicative bow-down trim, m	Initial heel, deg
half-liquids-all-slack-sensitivity	259.665	1.4207	4.1401	0.4960	1.0734	-0.467	0.00
half-liquid-port-crowding-sensitivity	259.665	1.4207	4.1401	0.4960	1.0734	-0.467	5.53
unverified-lightship-KG5.4-sensitivity	289.010	1.5474	4.5564	0.0000	0.8505	-0.812	0.00

Negative trim means stern-down. The full nominal schedule weighs 289.010 t, leaving **86.40053 t** unexplained relative to the retained design-waterline buoyancy. Its mass does not establish the actual lightship. Significant calculated trim requires a nonuniform-waterline hull solution; the table's initial GM and heel are upright, even-keel sensitivity outputs. A positive initial GM is not a completed passenger-ship stability assessment.

5. Verification and limits

Gauss-Legendre integration splits the hull at its geometry changes and waterline intersections. A transformed vertical coordinate removes the keel-section singularity. Orders 32, 64 and 128 produce displaced-volume agreement better than 0.000001 m³. Waterplane quantities converge separately, and the finite-difference displaced-weight derivative agrees with TPC. A separate direct-theta polygon-section calculation confirms volume within 0.04 m³; a direct parametric point test confirms the analytic inverse independently.

Ten numerical/regression tests pass. They cover original geometry equivalence, independent integration, convergence, unit consistency, density changes, conditional buoyancy roots, free-surface and CG arithmetic, negative GM/heel/trim flags, the impossible original aft peak, coordinated tank reserves and invalid numeric inputs. These checks establish calculation behaviour only. The game hull lacks evidence of actual plate offsets, sealed openings, appendage displacement, shell thickness, permeability and as-built weights.

The current local ferry-ship.js has also been checked against these results: published design displacement 375.41053 t differs from the calculated value by only 0.000000515 t; published Cb 0.74026634 agrees with the explicit LBP 31 m reference. The source retains 210 t lightship, 283 t operating target and the 165.41053 t arithmetic residual as unreconciled assumptions. It marks GM 1.6 m as game motion only, sets massReconciled:false, and withholds construction approval. Its relocated aft-peak identifier has z14.1-15.4 m and parent fwd-double-bottom, with a conformal shape and planning-capacity status. This verification concerns the local corrected source; it is not evidence of publication or a real ship approval. The exact checked fields are preserved under currentSourceVerification in the JSON.

6. Construction information still required

No plate thickness, beam section, weld size or engine-foundation bolt is invented or released by this review. A real drawing release requires the following controlled inputs and calculated documents:

- Confirmed service area, flag and applicable rules; assessed historical 1994 design basis separately from current reference guidance.

- Fair lines plan and offsets, watertight 3D hull, baseline/frame/perpendicular definitions, appendages and an independently checked hydrostatic dataset.

- Controlled structural and machinery weight estimate with VCG/LCG/TCG; hull and superstructure materials; completed lightship survey and inclining evidence at the applicable stage.

- Approved loading cases, KG limits, GZ curves and area checks, free surfaces, crowding/turning/wind, downflooding, trim, subdivision and damage stability.

- Scantling and global/local strength calculations: bottom/slamming and side/deck loads, vehicle concentrated/axle loads, bulkheads, tank hydrostatic loads, torsion, fatigue where applicable and buckling.

- Framing, shell/deck/bulkhead plating, connection/weld schedules and material specifications; fire and watertight boundary penetrations.

- Engine-bed local strength and fatigue, resilient-mount and bolt/chock design from vendor loads, lifting loads and overhaul paths, gearbox and thrust transmission.

- Shaft alignment and bearing reactions, shaft strength/critical speeds/torsional vibration, stern-tube and bracket/rudder/skeg construction, propeller/hull clearances and immersion in actual loading conditions.

- Tank boundaries and usable capacities including structural deductions; calibration/sounding tables; vents/overflow and internal access; corrosion/coating and pressure-test particulars.

- Freeboard/load-line and tonnage measurement, operating/loading limitations and flag/class plan disposition.

7. Primary reference evidence

These sources establish the scope of a genuine stability review. They do not certify this concept:

1. [IMO, Ship Design and Stability](https://www.imo.org/en/ourwork/safety/pages/shipdesignandstability-default.aspx): identifies subdivision, stability, free surfaces, watertight integrity and the relevant convention/code framework.

2. [IMO MSC.267(85), 2008 IS Code](https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.267%2885%29.pdf): Part B 3.1 on liquid free surfaces, chapter 8 on lightship parameters and annex 1 on inclining procedures.

3. [ClassNK Passenger Ship Rules, December 2025](https://www.classnk.or.jp/hp/pdf/rules/501_passenger_e_202512.pdf): Part 4 chapter 4 calls for service-condition stability and heeling curves, determination of lightship centres of gravity and stability information for the master. Its applicability to any hypothetical real vessel must be determined by the chosen flag/class design basis.

Final disposition: the calculated design-waterline particulars and relocated tank envelopes are coherent at concept level. Weight, trimmed hydrostatics, full stability, actual tank capacity, scantlings and plan approval remain open engineering holds; the drawings must not be marked “approved for construction”.

Machinery particulars and service calculations

Minato Maru - machinery design review M1

Issue purpose: preliminary engineering and game coordination. **Construction release:** withheld. **Reviewed basis:** current main snapshot e87c06b16734d882626be7fe6cdd2ab153d649c4, 13 October 2026. Equipment and arrangement reservations below must retain their qualification on drawing sheets.

The 1994 fictional Johansson Marine JM6-160 machinery remains the game identity. The physical reference is a later Japanese manufacturer package; it does not establish what a 1994 ferry actually carried. The installed-fiction fields and manufacturer-reference fields are separated in the data model to prevent that substitution becoming an undocumented assertion.

Engine particulars and checked arithmetic

Particular	Installed game design	Evidence/status
Main engines	2	Existing design
Power, each	460 kW at 1,350 rpm	Fictional design, not an OEM rating
Cylinders	6 inline, four-stroke	Existing design
Bore x stroke	160 x 200 mm	Fictional design
Swept volume, calculated	24.127 L	$\pi/4$ x bore² x stroke x cylinders
Brake mean effective pressure	16.947 bar	1,200 x kW / (rpm x litres)
Mean piston speed	9.00 m/s	2 x stroke x rpm / 60
Rated crankshaft torque	3,253.83 Nm	Power / angular speed
Fictional engine dry mass	3,000 kg	Unverified supplier mass
Gear ratio	3.00:1	Existing design, no capacity certificate
Propeller speed	450 rpm	Consistent with the game engine and ratio
Shaft torque, assumed 97% gear efficiency	9,468.66 Nm	Calculated screening value

These values describe a plausible engine class. Numerical consistency does not prove thermal capability, durability, emissions, noise, thrust capacity or suitability for the selected propeller. The original rounded 24 L and 17 bar may remain short labels; the engineering record must use the calculated values and their basis.

Manufacturer comparison, deliberately separate

The official Yanmar 2020 brochure identifies the mechanically controlled 6AYM-WST C-rating at 485 kW/1,900 rpm: six 155 x 180 mm cylinders, 20.379 L, 207 g/kWh at rated output, 60 L jacket-water capacity and 91 L lubricating-oil capacity. It lists a 24 V/8 kW electric starter with air starting optional. [S1]

The manufacturer-channel legacy WST datasheet gives the engine-only outline as 2,000 x 1,305 x 1,431 mm, with dry mass 2,365 kg. Its matched YX-181 hydraulic reversing gear weighs 560 kg and offers 2.08, 2.55 and 3.03 ratios. The illustrated package is 2,364 mm long; aggregate dry mass is 2,925 kg. This sheet calls the engine H-rating and lists a 7 kW starter, so that legacy hardware description must not be silently merged with the later C-rating configuration. [S2]

The reference package is attractive because it has mechanical control, individual-cylinder construction and published package geometry. Its 1,900 rpm rated speed is materially different from the game engine's 1,350 rpm. The reference 3.03 ratio yields approximately 627 rpm, rather than the game's 450 rpm. It is therefore a comparison candidate, not a confirmed replacement with the existing shaft line and propeller.

Mitsubishi's S6A3-MPTAW offers another documented physical comparison: 18.56 L, 1,636 x 1,036 x 1,421 mm and 1,900 kg. No complete matched gearbox interface was obtained. The S6A3-MPTA page includes generator-engine ratings; these must not be represented as approval for propulsion duty. [S3]

The gear offset, output coupling, foot coordinates, mounting bolts, flexible-coupling particulars and certified equipment centres of gravity remain **unknown**. Neither an outline sheet nor a flywheel-housing designation substitutes for installation drawings. The 2020 brochure's optional resilient mounts are shown for a freestanding gear arrangement;

they must not be copied into an integral gear installation without confirmation.

Dimensioned arrangement and removal reservations

Coordinates are metres: **x positive to port**, **y above keel**, **z forward from the transom**. A 2.70 m long x 1.50 m wide x 1.65 m high assembly reservation is centred at x = +/-1.55 m, z = 10.70 m, with lower edge y = 0.90 m. Each envelope spans z = 9.35-12.05 m. The port envelope occupies x = 0.80-2.30 m; starboard occupies x = -2.30--0.80 m. The envelope centre y = 1.725 m is a geometric planning point, **not certified equipment CoG**.

Reserve 3,600 kg per assembly: 3,000 kg fictional engine plus a 600 kg gearbox allowance. Foundations, shafting, insulation, liquids and lift rigging are additional. Actual weights and CoG must replace these allowances in the weight register before stability or foundation approval.

Planning access targets are 1.00 m on the operating side, 0.80 m opposite and 0.60 m at each longitudinal end. They are yard targets, not manufacturer or class minima. Existing nearby equipment, casing, control booth and generator geometry require a coordinated clash review. No relocation is authorised merely by drawing a rectangle in another already occupied room.

Current floor-plate elevation is 0.90 m and nominal underside of the deckhead is 2.80 m: clear height is only 1.90 m. The reserved machinery reaches 2.55 m, leaving 0.25 m overhead. That space cannot be certified for piston/liner lifting or a chain block. The 2.00 m clear-route planning target is not met before adding piping, cables and structure. This is an unresolved physical constraint, not a passing clearance calculation.

Draw 2.90 x 1.80 m clear removal-hatch reservations above each engine. The proposed clearance over the assembly outline is 100 mm at either end and 150 mm at either side. These are reserved openings only. A continuous vertical removal route, deck cut-outs and reinforcement, closure arrangements, lifting eyes, crane radius and vessel stability during the lift remain undemonstrated. The notional 3,600 kg assembly plus 400 kg rigging produces a 4,000 kg planning lift; **no lifting-appliance WLL or safe lift plan is selected**. Whole-engine extraction alongside with a shore crane is an option to investigate, not an established operation.

Fuel budget and corrected endurance claim

Use explicit duty assumptions: both engines at 80% of rating; assumed SFC 210 g/kWh; fuel density 0.86 kg/L; 22-minute crossing; 20 crossings/day. Assume 65 kW total generator electrical load and 260 g/kWhe. The generator figure is separate from engine shaft SFC and includes the chosen aggregate conversion assumption.

Calculated result	Value
Main-engine fuel use, both	154.56 kg/h
Generator fuel use	16.90 kg/h
Propulsion fuel per crossing	65.90 L
Crossing fuel including generators	73.10 L
Crossing-only daily budget	1,462.06 L/day
Usable storage: 15 m³ gross less 10%	13,500 L
Endurance on that crossing budget	9.23 days
Usable 1.5 m³ day tank autonomy	6.77 h

The former 46 L/crossing and sixteen-day tank endurance are not supported by this duty basis. Alongside/overnight operation, warm-up, diversions, reserve policies and weather are excluded; actual endurance may be shorter. No fuel-economy guarantee is implied. Service-tank volume also varies between the original arrangement prose (1.6 m³) and machinery particulars (1.5 m³); this must be normalised, not left as two different capacities.

Ventilation, cooling, shaft and starting-air screens

Ventilation assumes 100 kW total room sensible heat, ambient air density 1.18 kg/m³, specific heat 1.005 kJ/kg K and permissible rise 10 K. Heat-removal flow is 30,356.7 m³/h. At full propulsion rating, fuel use is 193.2 kg/h; assumed 32:1 combustion-air/fuel mass ratio adds 5,239.3 m³/h. The conservative sum is 35,596.0 m³/h. Reserve 40,000 m³/h for a supplier design exercise. At an assumed 750 Pa and 65% total efficiency, this needs 12.82 kW total electrical input, 6.41 kW per fan for two equal units. Existing two 3.7 kW motors cannot prove delivered flow. This calculation does **not** select a fan, verify a trunk or revise a cable rating; actual OEM heat/air figures and measured duct losses are missing.

Cooling assumes fuel LHV 42,700 kJ/kg and 30% of fuel input rejected to cooling water. Per main engine, input heat is 1,145.78 kW and assumed water heat is 343.735 kW. At seawater density 1,025 kg/m³, cp 3.99 kJ/kg K and 10 K rise, screening raw-water flow is 30.26 m³/h per engine. The value is neither the engine pump rating nor an approved pipe size. Separate jacket, charge-air, oil and gear cooler duties, tropical seawater limit, fouling, pressure losses and pump NPSH remain open.

The existing solid 110 mm shaft carries 36.23 MPa in a pure-torsion screen. With arbitrarily assumed 40 MPa screening allowable, torsion-only diameter is 106.43 mm. That narrow nominal comparison cannot establish adequacy: propeller bending, cyclic thrust, fatigue, material certificate, keyways, bearing span, alignment and torsional vibration are excluded. Retain the 110 mm diameter only as a game assumption pending a class shaft calculation.

Two 150 L receivers falling from 30 to 18 bar gauge provide an isothermal reference-air estimate of 3.60 m³ at 1 bar absolute and the same assumed temperature as the receivers. This is not a normal-cubic-metre specification. Manufacturer air consumption per start, temperature correction and required consecutive starts are unknown; the available engine-start count must remain null. The game's three-bar deduction per start is a gameplay rule, not a receiver-capacity proof. The original 2.94 MPa equals 29.4 bar rather than exactly 30 bar; drawings must distinguish design pressure, normal working pressure and nominal legacy kgf/cm² labels.

Review disposition

Closed by this review: arithmetic for displacement/BMEP/piston speed/torque; source separation; package-length identity; duty-defined fuel calculation; transparent screening formulas; explicit reservation versus actual installation data.

Construction holds: verified OEM engine/gear duty and period identity; certified weights/CoG; hull/shaft-line geometry and propeller matching; foundations and thrust reactions; fuel quality/pressures; cooling and sea-chest geometry; air-start capacity; ventilation fan/duct verification; exhaust backpressure, expansion and heat shielding; component overhaul access; vertical removal route and lift study; fire suppression protected volume and boundaries; shutdown/quick-closing valve arrangements. These holds remain outstanding and must be included in the final combined review.

Source code: src/world/ferry-machinery-design.js. Regression checks: tests/ferry-machinery-design.test.mjs. Machine-readable report: machinery-specs.json. Software checks validate the design record's calculations; they do not provide ship or equipment certification.

Sources

[S1] YANMAR, 6AYM-WST/WET manufacturer brochure, engine and marine-gear specifications; accessed 13 October 2026: <https://www.yanmar.com/media/news/2020/05/22051608/6aym-wstwet.pdf>.

[S2] YANMAR Norway, legacy 6AYM-WST datasheet, specification table and combined YX181 outline; accessed 13 October 2026: <https://www.yanmar.no/theme/yanmarportal/uploadedFiles/Marine/productDownloads/Commercial-datasheet/6148-Datasheet-6AYM-WST.pdf>.

[S3] Mitsubishi Turbocharger and Engine Europe, S6A3-MPTAW product particulars; accessed 13 October 2026: <https://engine-genset.mhi.com/marine-engine/s6a3-mptaw>.

Electrical correction and verification

Minato Maru electrical engineering review

Revision: corrective engineering package, 13 October 2026. Source baseline e87c06b16734d882626be7fe6cdd2ab153d649c4.

Status: engineering development - not for construction. Numeric defects are corrected and regression checked. Manufacturer selection, generator dynamics, actual cable installation, fault duties and protection coordination remain review holds. No class approval or physical plant validation is asserted.

Corrected transformer and shore schedules

The former lighting transformer record combined a 105 V, approximately 165 A secondary winding with the 440 V primary's 50 A breaker and 10 mm² cable. Each transformer now has separate primary and secondary records. Compatibility fields amps, breaker and cable describe the output winding. Frame and provisional long-time setting are distinct.

Source	Side	Voltage V	Rated A	Frame A	Setting A	Live cores x mm²	PE mm²	Planning cable rating A
T1	primary	440	39.4	50	40	3 x 10	10	50
T1	secondary	105	165	200	165	3 x 95	95	201
T2	primary	440	39.4	50	40	3 x 10	10	50
T2	secondary	105	165	200	165	3 x 95	95	201
ETR	primary	440	6.6	10	7	3 x 1.5	1.5	16
ETR	secondary	105	27.5	32	28	3 x 6	6	36
shore	primary	200	144.3	160	145	3 x 70	70	166
shore	secondary	440	65.6	80	66	3 x 25	25	89

Transformer line current is kVA x 1000 / (√3 x line-to-line voltage), assuming three-phase windings. The 105 V system supplies single-phase line-to-line circuits; secondary winding connection and phase balancing still require design. The ship shore connector receives 440 V from the quay's 200/440 V transformer. The former 75 A record did not identify its voltage side. New frames/settings are provisional selections, pending inrush, thermal, fault and time-current studies.

The main generators retain their 125 kVA/100 kW/445 V nameplates (162.2 A each) and 200 A frame/170 A game setting. Nominal distribution is explicitly 440 V; set-points and available kVA at that voltage require OEM confirmation. Both main generating sets are relocated to the auxiliary machinery room to clear the main-engine maintenance envelope. Two parallel 95 mm² sets per phase assume matched length and terminations. The space-centre route estimate to the control booth is 13 m. Each parallel set carries approximately 81.1 A; hot resistance-only generator-cable drop is 0.0964% at 445 V. Actual tray and transit routing remains unverified. The emergency generator remains 40 kVA/32 kW/445 V (51.9 A), with 63 A frame/52 A setting and 16 mm² live conductors plus PE. Those selections are not proof of generator thermal protection or starting capacity.

Cable calculation and complete inventory

Copper resistivity is ρ(T) = 0.0175 x [1 + 0.00393 x (T - 20)] Ω·mm²/m. At 90 °C, ρ = 0.02231425, 27.51% above its 20 °C value. The full-load model now uses that hot-conductor value. Three-phase drop uses √3 x I x ρ x L / S / U; single-phase and 24 V DC use 2 x I x ρ x L / S / U. L is one-way length, so the two-wire formula includes the return conductor. These resistance-only estimates exclude transformer regulation and complete upstream impedance.

The cable selector now fails explicitly if its largest listed section cannot meet ampacity and drop; it cannot silently accept an inadequate final size. A 5% final-feeder budget reserves a nominal 1% for upstream cables, but that upstream allocation remains unverified. All modelled feeders include a separate, conservatively full-size PE. Three-phase board feeders have three live cores plus PE; single-phase feeders have two live cores plus PE. Star-delta motor tails identify six live cores plus PE and must not be confused with three-core supply feeders. Actual tail length and section still require terminal, thermal and short-circuit verification.

Two retained room-air-conditioner feeders now serve all four units, each carrying two units at 22.4 A, with 25 A protection and 6 mm² cable. The three workshop motors and two toilet fans now retain their complete inventories. Motors use the higher of declared current and the provisional 90%-efficiency/0.85-power-factor calculation; this avoids reducing an unverified declared current. Actual nameplates remain required.

Ampacity is a planning lower envelope bounded by the referenced three-loaded-core, 90 °C insulation, 45 °C ambient rule table. It is not a certified cable manufacturer schedule. Grouping factor 1.00 is assumed; a 0.85 sensitivity is published. The thruster's 200 A feeder has no adequate grouped selection in the retained maximum 120 mm² range. This is a route/installation review hold, not a claim that the eventual route must use that factor.

Numeric checks cover 58 feeders. Maximum final-feeder hot-resistance drop: 4.7%. Maximum base-case motor-start cable envelope: 4.5%.

Declared operating-load scenarios

Condition	Declared kW	Estimated kVA	Source kW	Source kVA	Active reserve kW	Apparent reserve kVA
sea	40.1	45	100	125	59.9	80
arrival	110	127.5	200	250	90	122.5
loading	45.9	51.7	100	125	54.1	73.3
night	10.3	11.2	40	50	29.7	38.8
blackout	10	11.6	32	40	22	28.4

These are the game's declared diversity scenarios. Apparent-power estimates assume PF 0.85 for modelled motors, 1.00 for resistance heaters, and 0.90 for other loads. They do not establish worst-case simultaneous demand, loss-of-generator duty, load unbalance, losses or harmonic capacity. The connected apparent-load inventory totals approximately 300.8 kVA; that sum is not simultaneous demand.

Motor starting and emergency-power hold

Base locked-rotor assumption is 6 x running current for DOL; star connection draws one third of that DOL line current. The cable-only envelope uses hot resistance and explicit assumed reactance 0.12 Ω/km. Sections increase where needed to satisfy a 5% cable-only start target. It does not demonstrate source voltage dip, frequency recovery, acceleration or torque. Star-delta transition separately carries a DOL-current sensitivity and remains unverified.

Emergency motor started	Start A	Start kVA	Background kVA	Total kVA	Generator continuous kVA	5-7 x total kVA range	Status
steer-2	24	18.3	11	29.3	40	26.2-32.3	HOLD: generator transient study
fire-emergency	60.6	46.2	6.3	52.5	40	44.8-60.2	HOLD: generator transient study

The emergency fire-pump start exceeds the 40 kVA continuous rating under the simple scenario. Transient capability may exceed continuous rating, but it is unknown here. Obtain alternator reactances/excitation data, AVR and governor step-load response, engine response, motor locked-rotor current/torque and inertia. Establish whether staged load pickup, an approved reduced-current starter or a larger emergency set is needed. Rated current alone is not a start test.

Unresolved release holds

- **ELEC-01:** Real generator/transformer/motor/cable selection and certified installation data.

- **ELEC-02:** Prospective fault levels, breaker making/breaking duties, time-current discrimination and cable fault withstand.

- **ELEC-03:** Generator step-load response; emergency fire-pump scenario exceeds 40 kVA continuous rating; current-only sizing is insufficient.

- **ELEC-04:** Actual cable routes, installation/grouping, transformer regulation, unbalance and complete upstream-to-consumer voltage drop.

- **ELEC-05:** Battery Ah/endurance, DC load inventory, 105/24 V emergency converters and independent transition lighting.

- **ELEC-06:** Emergency duration against vessel-specific flag/class requirements; a 60 L/15 h claim lacks fuel-consumption evidence.

- **ELEC-07:** Motor six-terminal compatibility, star torque/acceleration, transition inrush and vendor packaged-equipment starting currents.

- **ELEC-08:** Switchboard construction, earthing/insulation monitoring, shore isolation, EMC and factory/onboard acceptance testing.

- **ELEC-09:** Mechanical ventilation duty and fan curves; fan motor kW does not prove airflow.

- **ELEC-10:** Navigation-light dedicated circuits, panel indications, changeover and lamp-voltage requirements; general feeder-drop budget is not navigation-light approval.

The 24 V loop formula is corrected, but the underlying game inventory lacks a complete DC load schedule. Individual emergency DC circuits, 105/24 V converters and battery capacity/endurance remain to be designed. ETR is a lighting transformer, not a battery charger. Steering feeders identify overload alarm without an overload trip; final protection requires equipment-specific review.

Primary reference and verification

[ClassNK Part H, June 2026](https://www.classnk.or.jp/hp/pdf/rules/420_part_h_e_202606.pdf), sections 1.1.6, 2.3, 2.8, 2.9 and 3.3; Table H2.15, printed page 34. Applicability to the intended domestic passenger/vehicle ferry must be agreed with flag and classification authorities. This is a reference basis, not an approval of the vessel.

Verification: node --test tests/ferry-electrical-engineering.test.mjs tests/ferry-kks-piping-electrical.test.mjs - nine tests passed, zero failed. Regression coverage: transformer side separation; hot-conductor and DC return formulas; all grouped consumers; PE and star-delta core counts; cable-only starting; emergency transient hold; grouping sensitivity. The relocation regression also verifies parallel-current sharing, route derivation and hot cable drop. Tests verify the software model and arithmetic, not a physical electrical plant.

Arrangement, routes and game access

Minato Maru machinery arrangement and piping review

Reviewed source baseline: e87c06b16734d882626be7fe6cdd2ab153d649c4. This review does not confer class, flag or construction approval.

Demonstrated corrections

Both generator sets have moved forward 2.65 m into the existing auxiliary machinery room. Their beds formerly occupied yard z12.55-13.40 m and now occupy z15.20-16.05 m. All generator geometry, maker plates and fuel/oil/water/air anchors follow that translation. Remote generator operation remains at the main switchboard; two local operating stations follow the auxiliary-room panels.

The main-engine air filter reaches yard z13.00 m. Relocating the generators removes the overlapping plan footprints at its forward end and leaves a 1.00 m service band to the engine-room forward bulkhead at z14.00 m. In the auxiliary room, the innermost generator panel faces are at x+/-0.50 m, giving a 1.00 m centre aisle. The outboard alternator features reach approximately x+/-2.445 m against side plant beginning at x+/-3.48 m, giving at least 1.035 m nominal side clearance. These are measured model clearances, not supplier overhaul minima. Model walks verify the centre aisle, both outer aisles and access to the local generator panels. The route crosses the auxiliary room orthogonally at z16.55 m, avoiding the sewage tank and generators.

The fuel quick-closing valve was placed at the deckhead and directed its 0.25 m terminal stub upwards into the car-deck structure. Its operating anchor has moved from 1.90 m to 1.50 m above the floor plates, with a 0.12 m stub. The outside-radius check places the terminal below the deckhead.

The watertight-door top frame projected 50 mm into the deckhead; its frame now finishes at the underside. The door notice and workshop sign were above the available 1.90 m room height; both are now on adjacent bulkhead panels below the deckhead. Door widths and room-to-room passage topology remain unchanged.

Engineering routing now checks every segment against conservative outside-radius-expanded equipment boxes and against already accepted independent pipe routes. It checks transverse, vertical and longitudinal machinery-room limits. Only a declared first/last terminal penetration may leave those limits. A first/last stub may enter a conceptual equipment box containing its recorded terminal; this does not establish physical nozzle ownership. Empty and nonfinite paths are rejected. Unsupported, missing and obstructed positive-DN edges produce explicit issues rather than assumed valid pipes. A bounded set of alternative lane positions/orders is attempted. Engineering segments retain source-line identity, so overlapping independent loops are not silently combined into a fabricated main.

The game retains its 172 illustrative paths, explicitly marked visual-preview-unverified, in FERRY_INTERIOR.pipes. Only FERRY_INTERIOR.engineeringPipes is the accepted engineering registry. The visual union of coincident preview bars and bends is solely a game rendering operation. The strict registry is cached on first access, keeping the approximately 1.3-second audit out of ordinary game startup. A measured standalone module import took about94 ms in this environment.

Actual arrangement and open holds

The floor remains0.90 m above the keel and the deckhead2.80 m above the keel:1.90 m nominal clear height before services. The engine-room plan remains yardz8.50-14.00 m andx+/-3.98 m. No deck elevation or ship stability property has been altered by this geometry work.

The main-engine guard rails leave approximately1.26 m between their inside faces in the centre aisle. This is a game collision measurement, not a supplier or statutory access approval. The primary ladder is0.75 m wide. The decorative aft starboard escape ladder is only0.40 m wide; no complete protected trunk, openable hatch and operating escape arrangement has been verified.

The lagged main exhaust run has a nominal outer top1.89 m against a1.90 m deckhead, leaving only10 mm nominal model clearance. Supports, insulation tolerances and hot-surface separation remain unverified. The restored forward service band does not establish a complete engine-overhaul procedure.

The current diagram-node anchors often use one position and one leaving direction for both device inlet and outlet. Fuel supply/return paths, serial valves and independent engine coolant/oil loops need separate physical ports and explicit manifold junction/spool records. Additional candidate paths cannot establish the correctness of an unknown port arrangement.

The current strict bounded result is55 clear paths and151 explicit issues across206 positive-DN diagram edges:9 missing physical terminals,25 systems without coordinated3D routing(exhaust and hydraulics),94 pipe-clearance candidate failures,18 equipment-clearance candidate failures and5 machinery-space-envelope failures. These are algorithm results, not proof that no feasible physical design exists. A complete piping redesign remains necessary. The JSON records every accepted edge, issue and current anchor coordinate.

No engineered removable hatch, lifting beam, rigging envelope, foundations, vibration isolation, flexible connection specification, hot-surface separation, pipe supports or watertight penetration schedule is validated. The check is against conceptual equipment boxes and a rectangular machinery-space envelope; it is not a complete closed-hull plating or supplier-nozzle clash approval. FERRY_ENGINEERING_ARRANGEMENT exports these holds for drawings and review.

Verification

Thirteen new regressions cover rejected-route handling, finite outside radius, terminal equipment-entry scope, successful alternative routing, independent-loop identity, all-edge coverage including unsupported systems, revalidation of the accepted registry, preview preservation, relocated generator footprints/labels/ports, local operating positions, all three auxiliary aisles, invalid-path rejection and longitudinal envelope limits.

The six-file scoped run comprises48 tests covering these changes plus existing ship geometry, machinery operation, game pipe preview, ladder access and below-deck walking. The unchanged render budget is100,000 triangles and fewer than40 meshes; the corrected model measures92,316 triangles in39 meshes. Passing game software checks does not release the vessel for construction.

Validation evidence and file use

Scoped source checks: 80 passed; 0 failed. These checks cover the retained ferry behaviour and new numerical, electrical and geometric regressions. They do not constitute full town/browser or live deployment certification.

Read the CAD drawing book alongside this report. Exact identifiers, circuits, node connectivity, accepted physical paths and unresolved path records are supplied in the CSV/JSON schedules. Proposed lift and supplier envelopes are reservations; they are not bolt patterns, certified crane ratings, pipe spools or approved foundation drawings.

Construction release requirements

- Confirm actual flag, service area, passenger/vehicle programme and applicable rule editions.

- Reconcile measured or calculated lightship weights, longitudinal and vertical centres, trimmed load cases, full righting-arm curves, damage stability, downflooding and propeller immersion.

- Complete scantlings, weld procedures, engine foundations, shaft alignment, bearing loads, thrust and torsional analyses.

- Obtain controlled manufacturer installation drawings, machinery duty approval, connection coordinates, pump curves, cooling and ventilation performance.

- Resolve every physical pipe route and define pipe classes, pressure/temperature limits, joints, supports, isolation, fire/watertight penetrations and test procedures.

- Validate fire boundaries, actual protected volumes, escape routes, extinguishing distribution and release/interlock functions.

- Complete generator transients, short-circuit levels, protection selectivity, installed cable grouping and certified lifting/removal paths.

- Obtain required design approval and construction surveys before an approved-for-construction issue.